Ex.No:5 Experiment on Packet capture tool: Wireshark.

Aim:

To determine the packet capture tool like wireshark.

Packet Sniffer:

* Sniffs messages being sent/received from/by your computer.

* Passive program,

⇒ never sends packet itself.

⇒ no packets addressed to it.

⇒ receives a copy of all packets.

Description:

Wireshark:

* Wireshark, a network analysis tool formerly known as ethereal, captures packets in real time and display them in human readable format. Wireshark includes filters, color coding and other features that let you dig deep into network traffic and inspect individual packets.

Wireshark used for,

⇒ Network administrators: troubleshoot network problems.

⇒ Network security engineers: examine security problems.

⇒ Developers: debug protocol implementations.

⇒ People: learn network protocol internals.

Getting Wireshark:

Wireshark can be downloaded for windows or macOS from its official website. For linux or another UNIX-like system, wireshark will see found in its package repositories.

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Capturing packets:

After downloading and installing wireshark, launch it and double-click the name of a network interface under capture to start capturing packets on that interface. As soon as click the interface's name, you'll see the packets start to appear in real time.

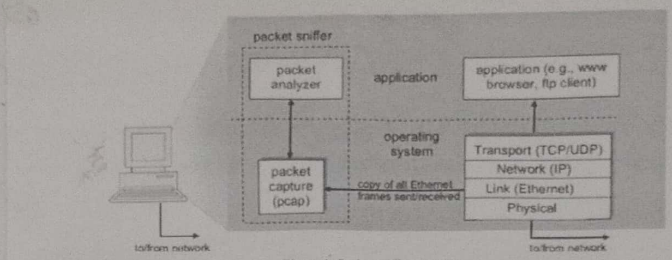
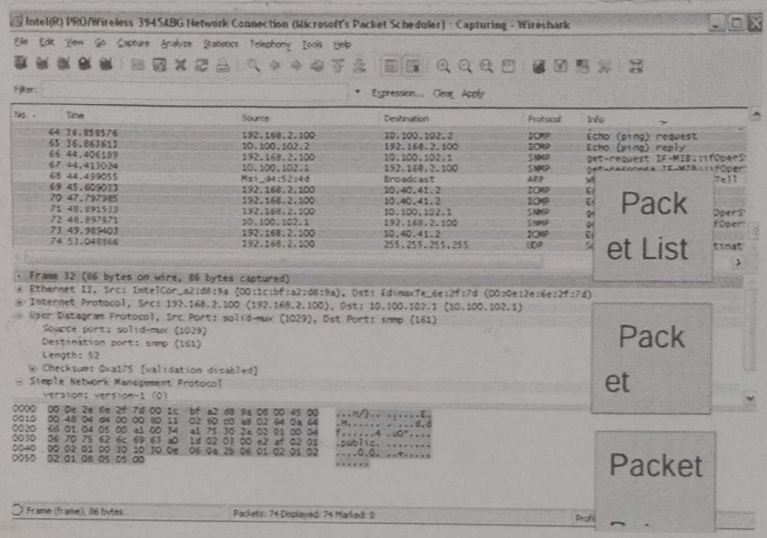


Figure 1: Packet sniffer structure



The "Packet List" Pane:

The packet list pane displays all the packets in the current capture file. The Packet list plane each line in the packet list corresponds to one packet in the capture file.

The "Packet details" pane:

The packet details pane shows the current packet in a more detailed form. The pane shows the protocol and protocol fields of the packet selected in the "Packet List" pane.

The "Packet bytes" Pane:

The packet bytes pane shows the data of the current packet in a hexdump style.

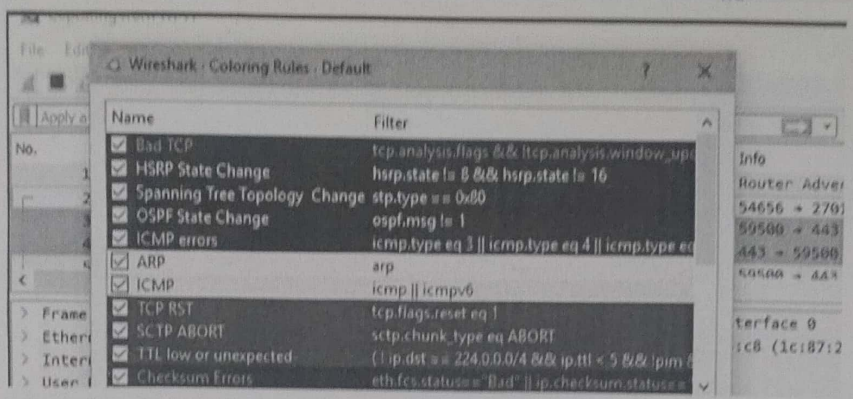
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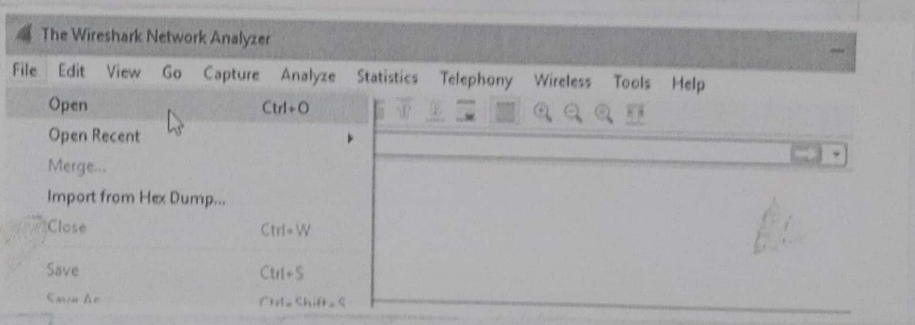
Color coding:

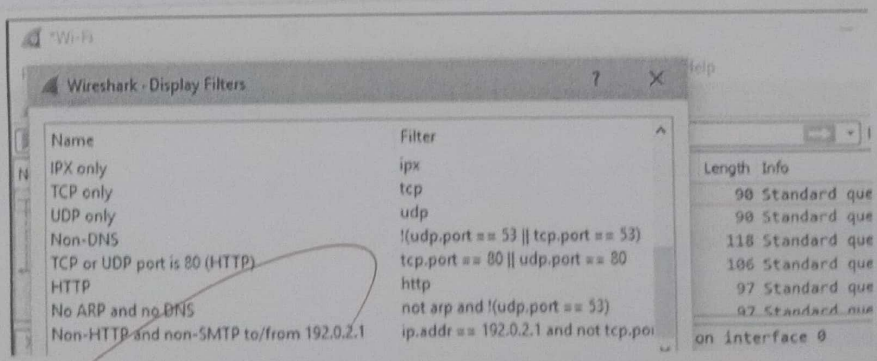
You will probably see packets highlighted in a variety of different colors. Wireshark uses colors to help you identify the types of traffic at a glance. To view exactly what the color codes mean, click View > Coloring Rules.

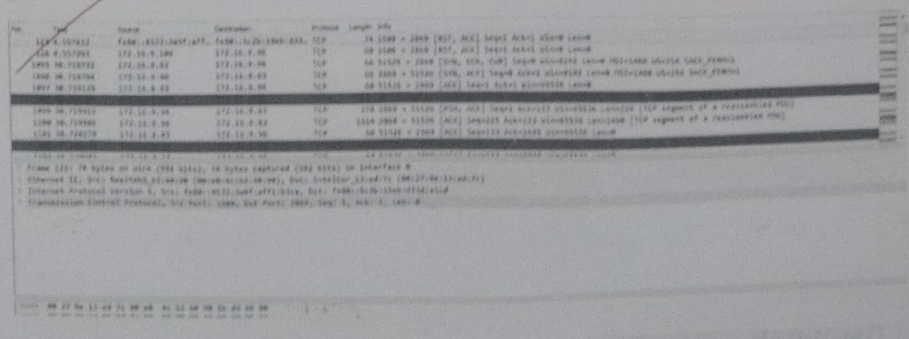
Filtering packets:

If you're trying to inspect something specific, such as the traffic a program sends when phoning home, it helps to close down all other applications using the network so you can narrow down the traffic.









Capturing and analysing packets using wireshark tool.

To filter, capture, view, packets in wireshark tool.

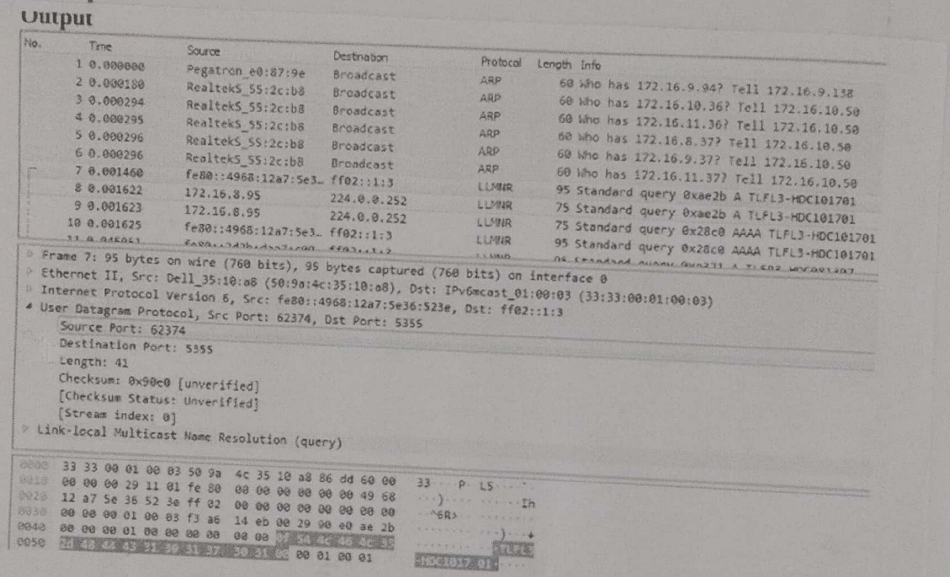
Capture 100 packets from the ethernet: IEEE 802.3 LAN interface and save it.

Procedure:

* Select local area connection in wireshark.
* Go to capture option.
* Select stop capture automatically after 100 packets
* Then click start capture.
* Save the packets.

Output:

Output



i) Create a filter to display only TCP/UDP packets, inspe the packets and provide the flow graph.

ii) Create a filter to display only ARP packets and inspect the packets.

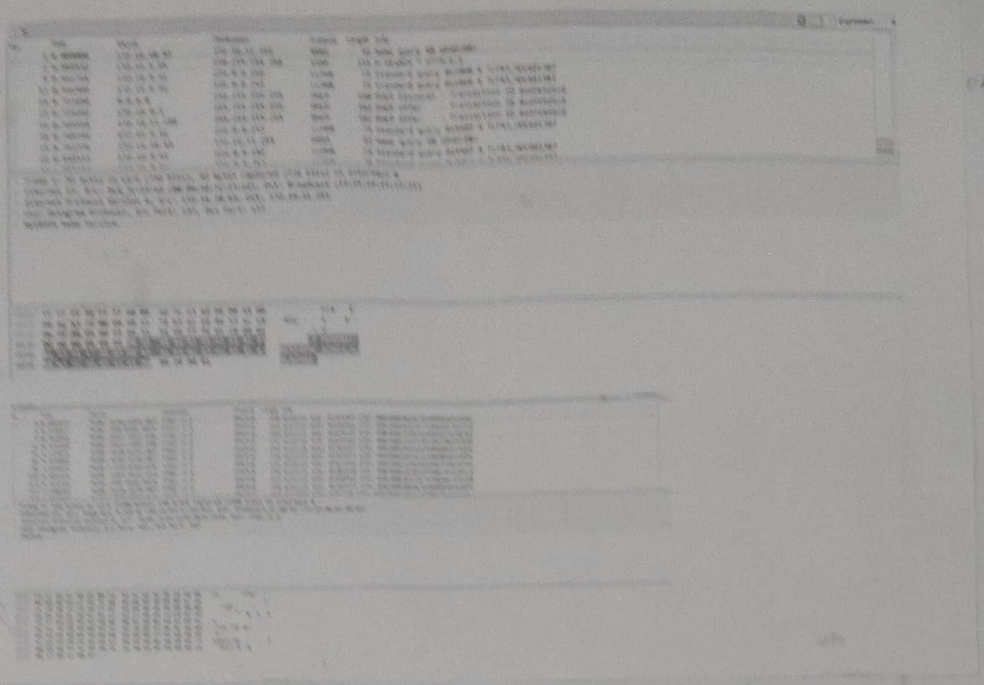
iii) Create a filter to display only DNS packets and provide the flow graph.

iv) Create a filter to display only HTTP packets and inspect the packets.

v) Create a filter to display only IP/ICMP packets and inspect the packets.

vi) Create a filter to display only DHCP packets and inspect the packets.

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Student observation:

1) What is promiscuous mode?

⇒ It is a network interface mode in which a device captures and receives all network packets.

2) Does ARP packet have a transport layer header? Explain

⇒ No, ARP works directly at the network access and is not carried using TCP or UDP.

3) Which transport layer protocol is used by DNS?

⇒ DNS primarily uses UDP at port 53 for normal queries. It can also use TCP port 53, especially for zone transfers.

4) What is the port number used by HTTP protocol?

⇒ The port number for HTTP is 80.

5) What is a broadcast IP address?

⇒ A broadcast IP address is an IP address used to send a packet to all devices on a particular network/subnet.

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